Amazon Food Reviews Analysis Project

本项目用于分析亚马逊食物评论数据，包括评分分布、情感分析与关键词提取

In [3]:
```python
import pandas as pd

df = pd.read_csv("../data/Reviews.csv")
print("找到数据： ",df.head())
print("分析总评论数:",df.shape)
print("字段类型:",df.isnull().sum())
```

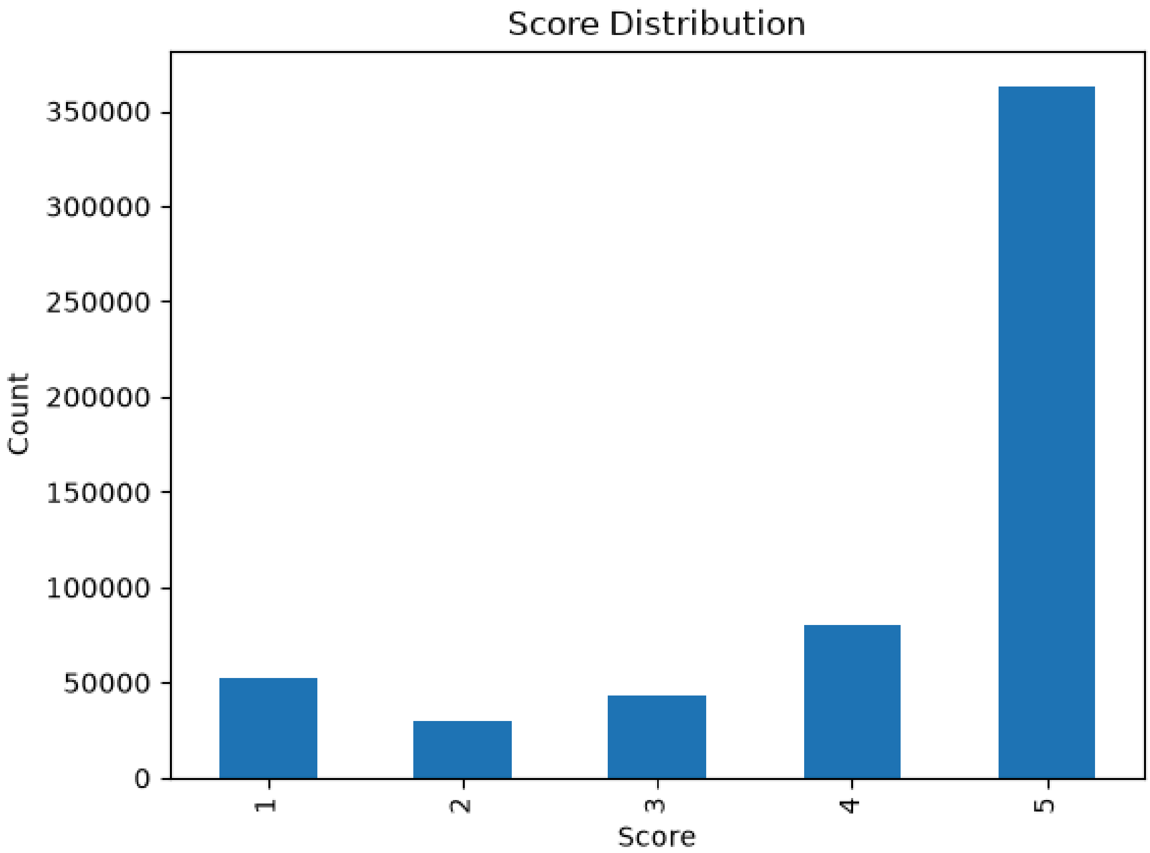
```
找到数据：      Id   ProductId          UserId                      ProfileName  \
0   1  B001E4KFG0  A3SGXH7AUHU8GW                       delmartian
1   2  B00813GRG4  A1D87F6ZCVE5NK                           dll pa
2   3  B000LQOCH0   ABXLMWJIXXAIN  Natalia Corres "Natalia Corres"
3   4  B000UA0QIQ  A395BORC6FGVXV                             Karl
4   5  B006K2ZZ7K  A1UQRSCLF8GW1T    Michael D. Bigham "M. Wassir"

   HelpfulnessNumerator  HelpfulnessDenominator  Score        Time  \
0                     1                       1      5  1303862400
1                     0                       0      1  1346976000
2                     1                       1      4  1219017600
3                     3                       3      2  1307923200
4                     0                       0      5  1350777600

                 Summary                                               Text
0  Good Quality Dog Food  I have bought several of the Vitality canned d...
1      Not as Advertised  Product arrived labeled as Jumbo Salted Peanut...
2  "Delight" says it all  This is a confection that has been around a fe...
3         Cough Medicine  If you are looking for the secret ingredient i...
4            Great taffy  Great taffy at a great price.  There was a wid...
分析总评论数: (568454, 10)
字段类型: Id                         0
ProductId                  0
UserId                     0
ProfileName               26
HelpfulnessNumerator       0
HelpfulnessDenominator     0
Score                      0
Time                       0
Summary                   27
Text                       0
dtype: int64
```

In [4]:
```python
df = df.dropna(subset=["Text"])
df = df.drop_duplicates()
```

In [5]:
```python
import matplotlib.pyplot as plt #画图
import matplotlib.pyplot as plt #可视化

rating_count = df["Score"].value_counts()
print("统计各评分数量:",rating_count)

rating_count.sort_index().plot(kind = "bar")
plt.title("Score Distribution")
plt.xlabel("Score") #情感
```

```
plt.ylabel("Count")

plt.show()
```

```
统计各评分数量: Score
5    363122
4     80655
1     52268
3     42640
2     29769
Name: count, dtype: int64
```

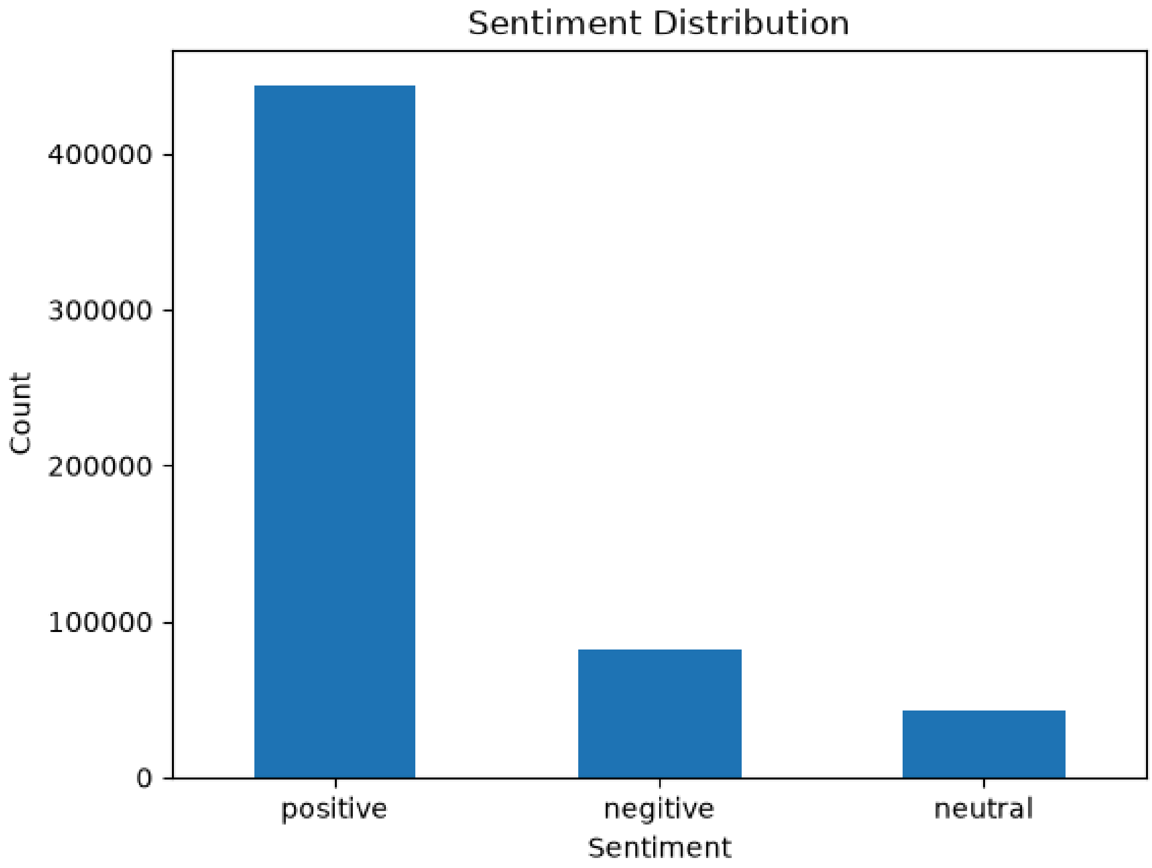


In [6]:
```
positive_rate = (
    len(df[df["Score"] >= 4]) / len(df)
)
print("用户满意度:",positive_rate)
```

```
用户满意度: 0.7806735461444549
```

In [7]:
```
def sentiment_label(x):
    if x >= 4:
        return 'positive'
    elif x == 3:
        return 'neutral'
    else:
        return 'negitive'
df["sentiment"] = df["Score"].apply(sentiment_label)
df["sentiment"].value_counts().plot(kind='bar')

plt.title("Sentiment Distribution")
plt.xlabel("Sentiment") #情感
plt.ylabel("Count")
plt.xticks(rotation=360)
```

```python
plt.show()
```

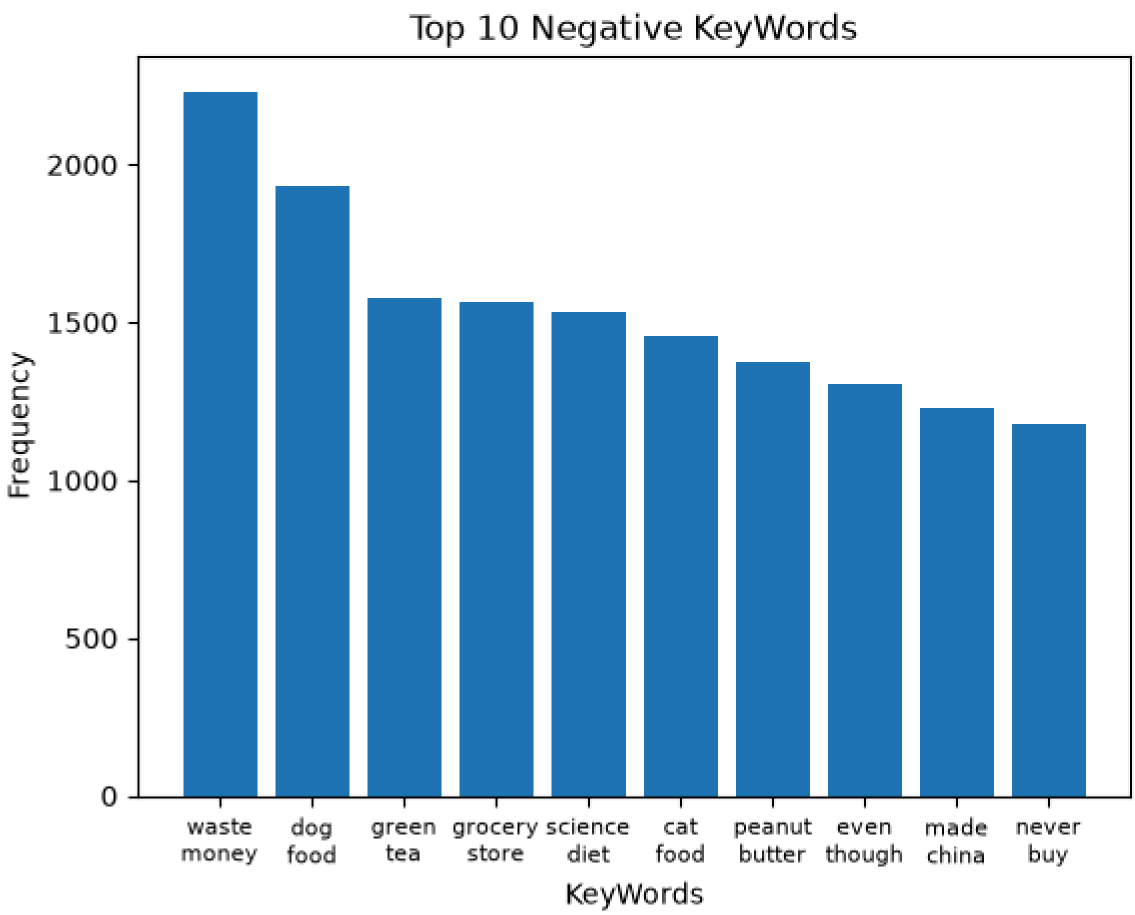


In [8]:
```python
print("统计情感分布:",df["sentiment"].value_counts())
print("评分 vs 情感关系:",df.groupby('sentiment')['Score'].mean())
```

```
统计情感分布: sentiment
positive     443777
negitive      82037
neutral       42640
Name: count, dtype: int64
评分 vs 情感关系: sentiment
negitive     1.362873
neutral      3.000000
positive     4.818253
Name: Score, dtype: float64
```

In [9]:
```python
import re # 数据清洗 NLP
# 去停用词 也就是主谓宾词
import nltk
nltk.download('stopwords',quiet=True)
from nltk.corpus import stopwords
# 定义两个组合在一起形成的差评词
from nltk.util import ngrams
from collections import Counter # 查出差评高频关键词

stop_words = set(stopwords.words('english'))
bad_review = df[df['Score'] <= 2]['Text'].fillna('').astype(str)
bad_review_text = " ".join(
    bad_review.apply(
        lambda x: re.sub(r'[^\w\s]','',x.lower())
    )
)
```

```
bad_words = bad_review_text.split()

clean_words = [
    w for w in bad_words
    if w not in stop_words
    and len(w) > 2
    and w.isalpha()
]
#筛选掉 由 not good 两个字组成的差评词 加入这个评论数据里面差评常常出现的高频谓语
bad_words_gl = {
    'would','could','should','know','think','thought','like','first','time','muc
    'also'
}
Zh_bigrams = list(ngrams(clean_words,2))
filter_bigrams = [
    bg for bg in Zh_bigrams
    if bg[0] not in bad_words_gl
    and bg[1] not in bad_words_gl
]
#让差评高频词更加人性化
badReview_top10_num = Counter(filter_bigrams).most_common(10)
#差评图
keywordsList = [' '.join(x[0]).replace(' ','\n') for x in badReview_top10_num]
badReview_top10Num_Scores = [x[1] for x in badReview_top10_num]

# plt.figure(figsize=(8,4))
plt.bar(keywordsList,badReview_top10Num_Scores)
plt.title('Top 10 Negative KeyWords')
plt.xlabel('KeyWords')
plt.ylabel('Frequency')

plt.xticks(fontsize=8)
plt.show()
```



In [10]:
```
# 评分 vs 情感关系
print("评分 vs 情感关系:",pd.crosstab(df["Score"],df["sentiment"]))
```

```
评分 vs 情感关系: sentiment  negitive  neutral  positive
Score
1               52268        0         0
2               29769        0         0
3                   0    42640         0
4                   0        0     80655
5                   0        0    363122
```

结论

整体好感度还是可以，但是产品质量与口感需要升级一下，尤其是宠物粮食这一块